

CodeCompass (Navigate Your Tech Career)

Overview

CodeCompass is an AI-driven career guide and personalized learning platform built for Computer Science students in the Philippines. It is a multi-platform product spanning a native Android app, a React web app, and a shared Django REST + Channels backend, delivering interactive learning roadmaps, gamified progress tracking, an AI career coach, a resume builder, and live job and certification discovery — all unified under a single account so students can move seamlessly between mobile and web.

Features

AI Career Coaching

- AI Chat Hub with multi-session history grouped by date (Today, Yesterday, Previous 7 Days)
 - General Career Advice for open-ended mentoring conversations
 - Roadmap Coaching for guided edits and progression help
 - Job Search Assistant for tailored search strategies
 - Resume Help for content critique and rewrites
 - Interview Prep for question banks and mock answers
- Real-Time Streaming Responses powered by Groq LLM over WebSocket
- Auto-Generated Session Titles from the first user message, pushed live via WebSocket events
- Conversational Onboarding Flow that collects year level, program, career goal, and interests through dialogue instead of forms

Personalized Learning Roadmaps

- AI-Generated Roadmaps tailored to the student's goal, program, and interests
- Node Types covering skills, projects, assessments, and certifications with difficulty, estimated hours, and linked YouTube resources
- Progress Tracking with in-progress and completed states and an automatic progress bar
- "Next Up" Banner that surfaces the next available node after each completion
- In-App Node Editor to add, rename, or remove nodes without regenerating the roadmap

- AI-Proposed Roadmap Switches and Upskill Add-Ons surfaced from chat sessions

Video Learning & Quizzes

- Embedded YouTube Player tied directly to roadmap node resources
- Auto-Generated Node Quizzes from video transcripts via the YouTube Transcript API
- Quiz Gating that unlocks XP and the next node only after passing
- Re-Watch Support so completed nodes can be revisited without re-gating the quiz

Resume Builder

- Multi-Section Editor for personal info, experience, education, skills, projects, and certifications
- Three Templates: Modern (two-column), Classic (serif), and Minimal (clean single-column)
- Live Preview that updates as you type
- AI Tools: bullet point generation from job titles, professional summary writer, job description parser, and ATS compatibility scoring with suggestions
- PDF Export through the browser print dialog

Jobs

- Live Job Listings aggregated from Careerjet (PH locale) and JSearch
- Search, Filters, and Save/Unsave actions
- Resume-Based AI Recommendations matched against career goal and parsed resume content

Certifications

- Catalog Browser for TESDA, Google, AWS, and other industry certifications
- Status Tracking with in-progress and completed states and completion dates

Universities

- Philippine University Catalog focused on CCS programs
- CHED CoE/CoD Badges for Centers of Excellence and Development
- Match Scoring against the student's profile and goal

Gamification & Achievements

- XP Points and Level Progression awarded for completed nodes, quizzes, and milestones
- Daily Streak Counter to reward consistent learning
- Badge Collection for milestones and category achievements
- XP History Timeline showing every awarded event
- Leaderboard ranked by total XP

User Management

- Email and Password Registration with email verification via Brevo HTTP API
- Google OAuth Sign-In on both Android (Credentials API) and Web (@react-oauth/google)
- Forgot Password and Reset Password flows
- Secure JWT Sessions with refresh-token rotation via SimpleJWT
- First-Login Onboarding gate before reaching the dashboard
- Profile Management with edit personal info, change password, and full account deletion

UI/UX Design

Design Principles

- Multi-Platform Parity: Consistent flows and visual identity across Android and Web
- Material Design 3 on Android: Native components, motion, and theming
- Modern Web Standards on Web: React 18, Vite, and Tailwind CSS utility-first styling
- Accessibility First: Semantic structure, keyboard navigation, and proper contrast
- Responsive Layout: Adapts from mobile devices to large desktop displays

Theme System

- Compass-Inspired Branding: Color palette aligned with navigation and discovery
- Custom Typography: Optimized for long roadmap descriptions and chat readability
- Consistent Spacing: Unified design tokens across web and Android

Security & Privacy

Data Protection

- JWT Access and Refresh Tokens via `django-rest-framework-simplejwt`
- `EncryptedSharedPreferences` on Android for secure token storage via `AndroidX Security Crypto`
- Token Refresh Authenticator that transparently renews expired sessions
- Email Verification before granting full account access
- Authentication Required on every roadmap, chat, resume, and job endpoint
- Rate Limiting via `django-ratelimit` on sensitive endpoints
- Input Validation on both client and server with `Zod` schemas on the web
- HTTPS/TLS for all REST and WebSocket traffic

Privacy Features

- Minimal Data Collection limited to essential profile, learning, and resume fields
- Per-User Data Isolation across roadmaps, chat sessions, resumes, and saved jobs
- Account Deletion that permanently removes the user and all related records

Performance Optimizations

Efficiency Features

- WebSocket Streaming for chat and onboarding to avoid polling
- Redis Caching and Channels Layer for real-time message delivery
- Celery + Beat Workers for background tasks like roadmap generation and quiz creation
- MVVM + Repository Pattern on Android with Retrofit + OkHttp connection pooling
- Vite Code Splitting on Web for fast initial loads
- Lazy Loading of charts, PDF parsing, and heavy components on demand

Resource Management

- `PDFBox Android` for in-app resume PDF parsing without server round-trips
- `pdfjs-dist` on Web for in-browser PDF text extraction
- Neon PostgreSQL for serverless, elastically scaled storage
- `Whitenoise` for efficient static asset serving on Render

Platform Support

Android

- Minimum SDK 31 (Android 12), Target SDK 36, Java 11
- AndroidX AppCompat, Activity, ConstraintLayout, RecyclerView, SwipeRefreshLayout
- Material Design 3, SplashScreen, Chrome Custom Tabs

Web

- Browser: Chrome 90+, Firefox 88+, Safari 14+, or Edge 90+
- JavaScript Enabled and a stable internet connection
- 1280×720 or larger recommended for the resume builder and dashboard charts

Acknowledgments

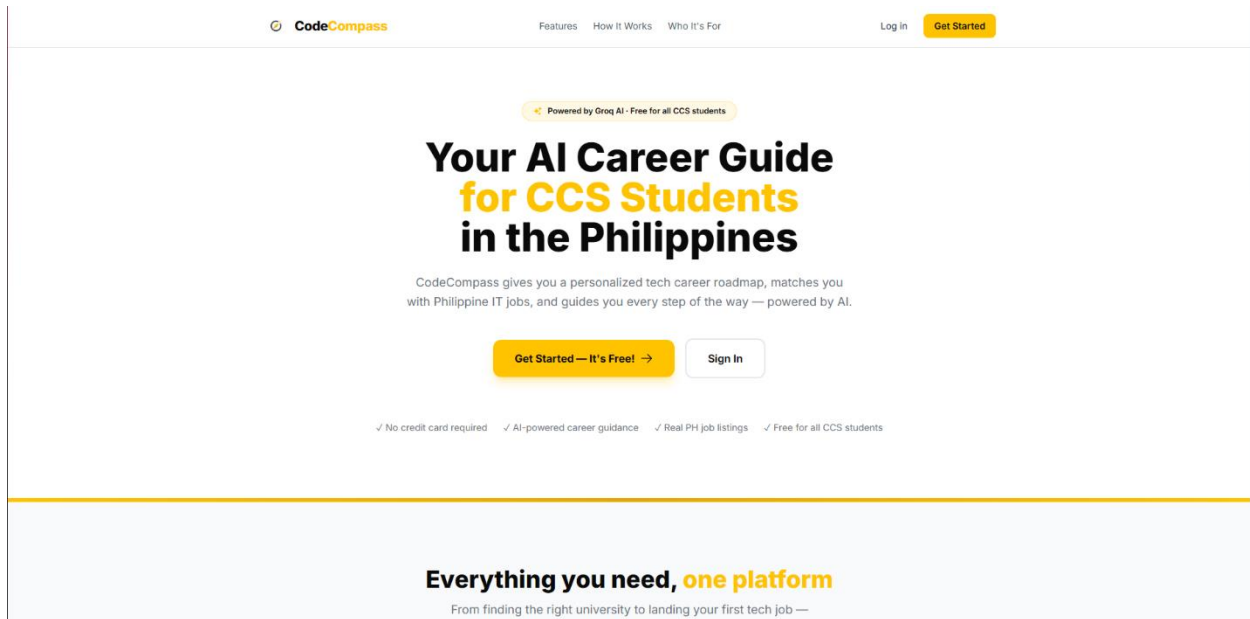
Technologies Used

- Android — Java 11, AndroidX, Material 3, Retrofit 2, OkHttp 4, Gson, AndroidX Credentials, AndroidX Security Crypto, PDFBox Android
- Web — React 18, Vite 6, React Router 6, Zustand, React Hook Form, Zod, TailwindCSS, @xyflow/react (roadmap canvas), Recharts, react-markdown, react-syntax-highlighter, pdfjs-dist, react-to-print
- Backend — Django 5.1, Django REST Framework, Django Channels 4, Daphne, SimpleJWT, django-cors-headers, django-filter, django-ratelimit,.djangorestframework-camel-case
- Database — Neon PostgreSQL via psycopg2 and dj-database-url
- Real-Time — Channels Redis, Redis 5
- AI — Groq SDK for streaming LLM responses
- Job APIs — Careerjet official Python client, JSearch
- Video — YouTube Data API v3, youtube-transcript-api
- Email — Brevo HTTP API via django-anymail
- Auth — google-auth (ID token verification), Google OAuth 2.0
- Tasks — Celery 5, django-celery-beat

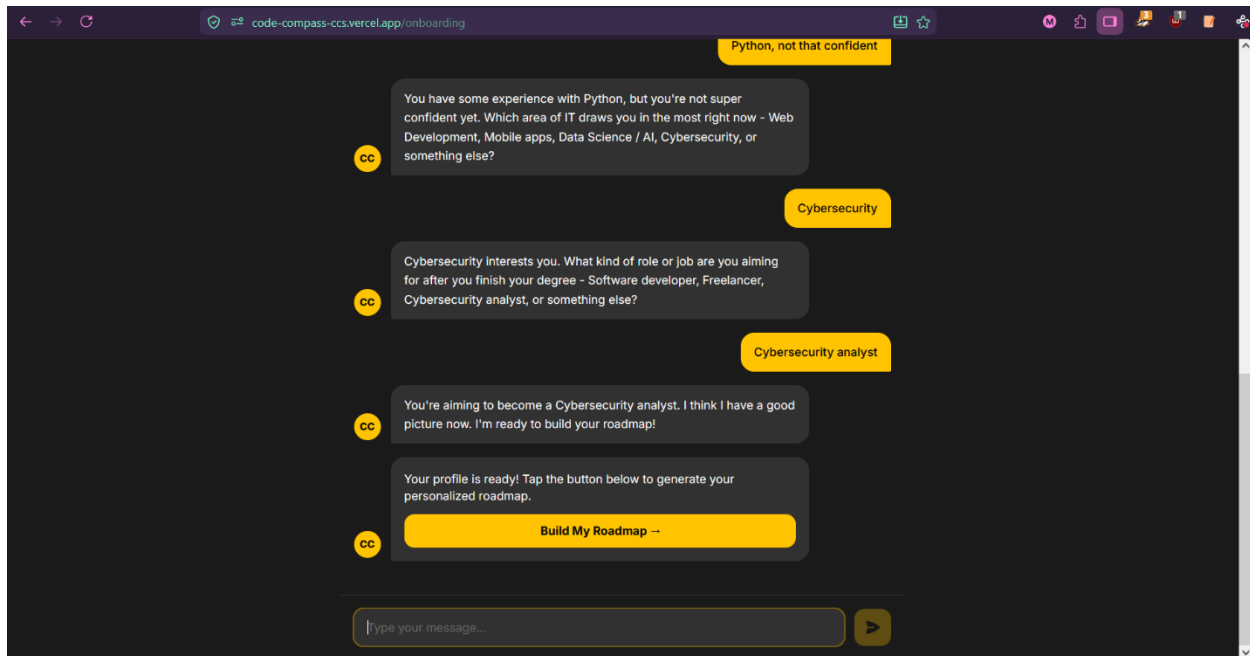
- Hosting — Render (backend, Daphne ASGI), Vercel (web), GitHub Actions CI for Android lint and APK builds

PAGE OVERVIEW

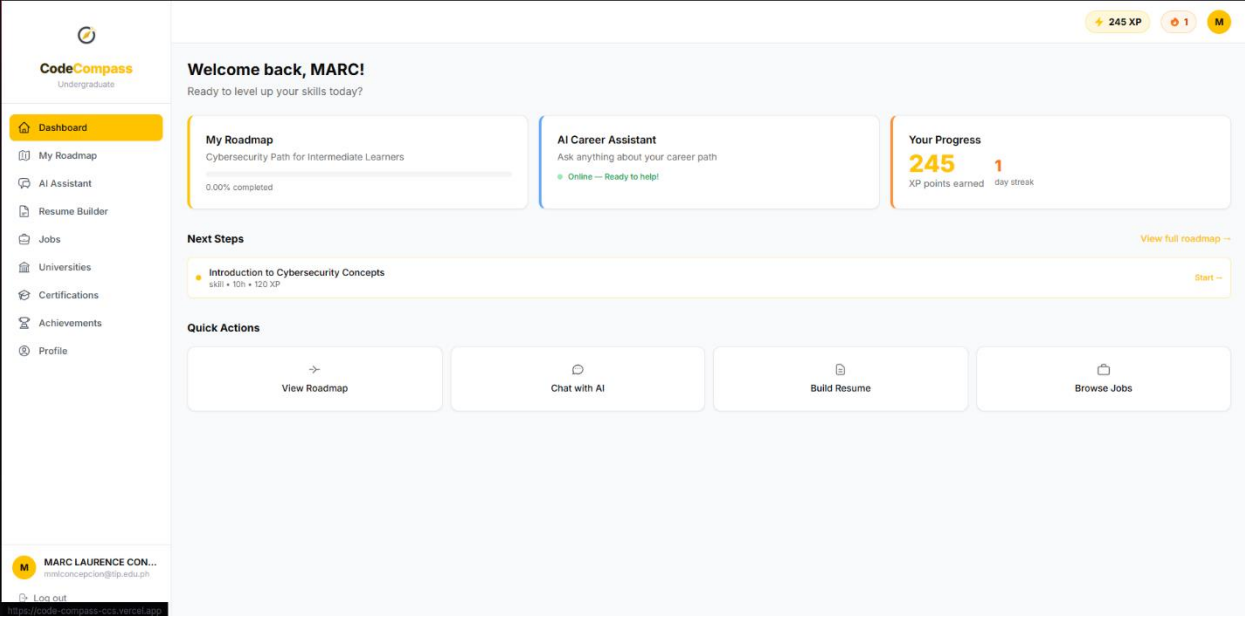
WEB



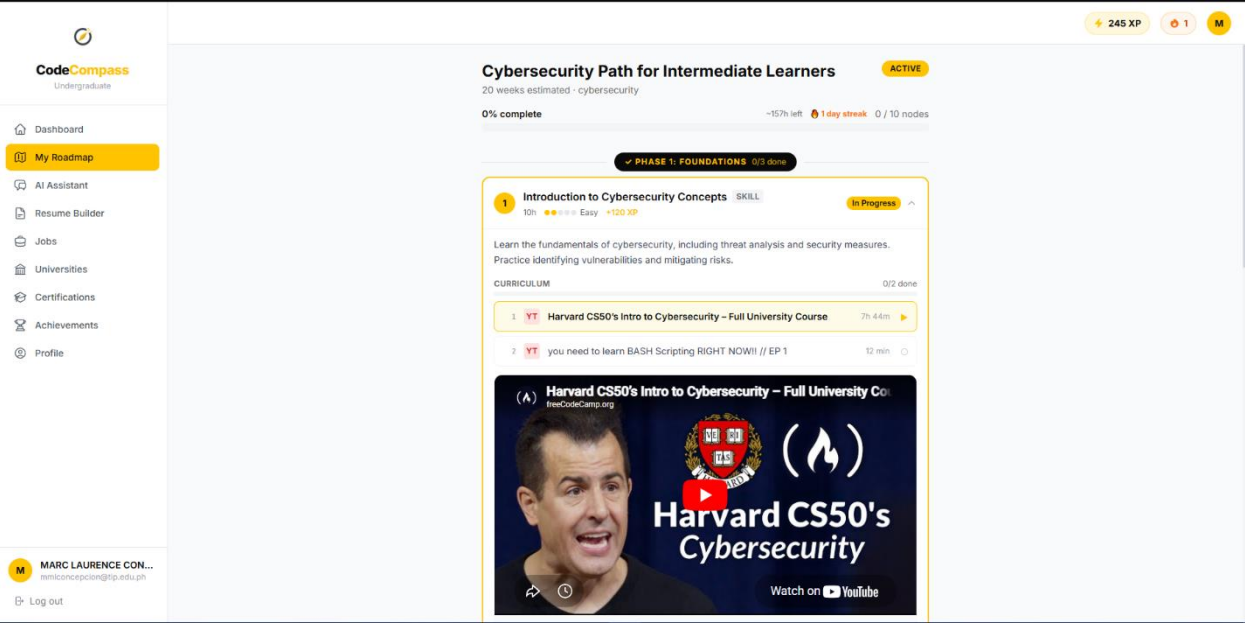
HOME PAGE



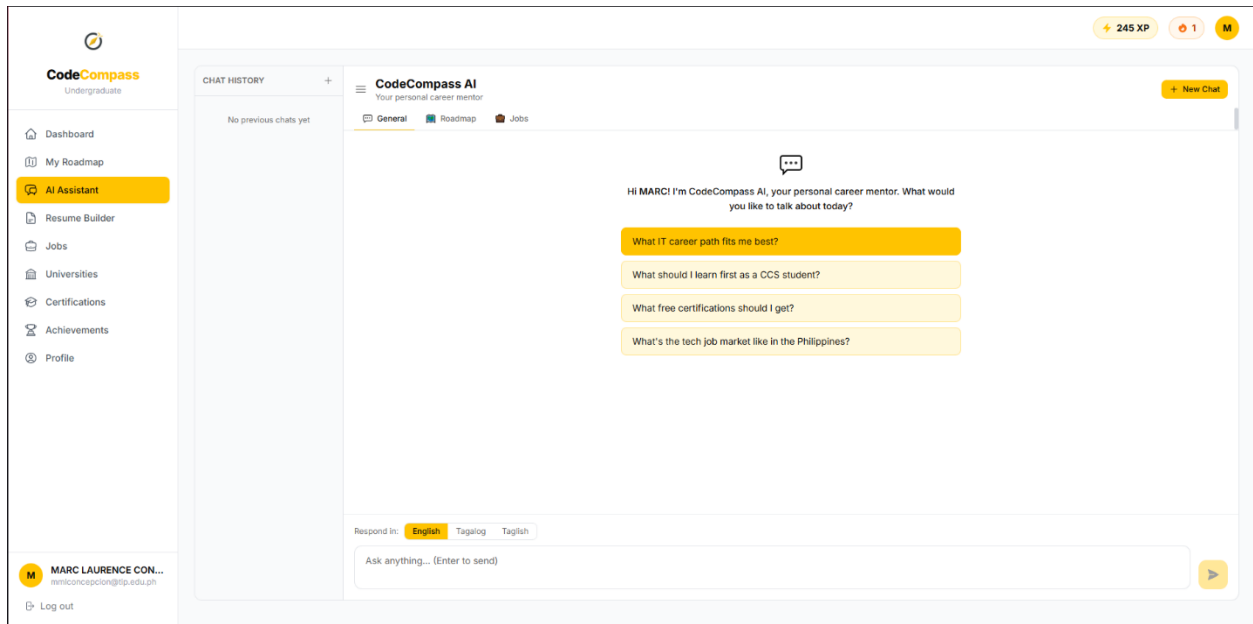
ONBOARDING



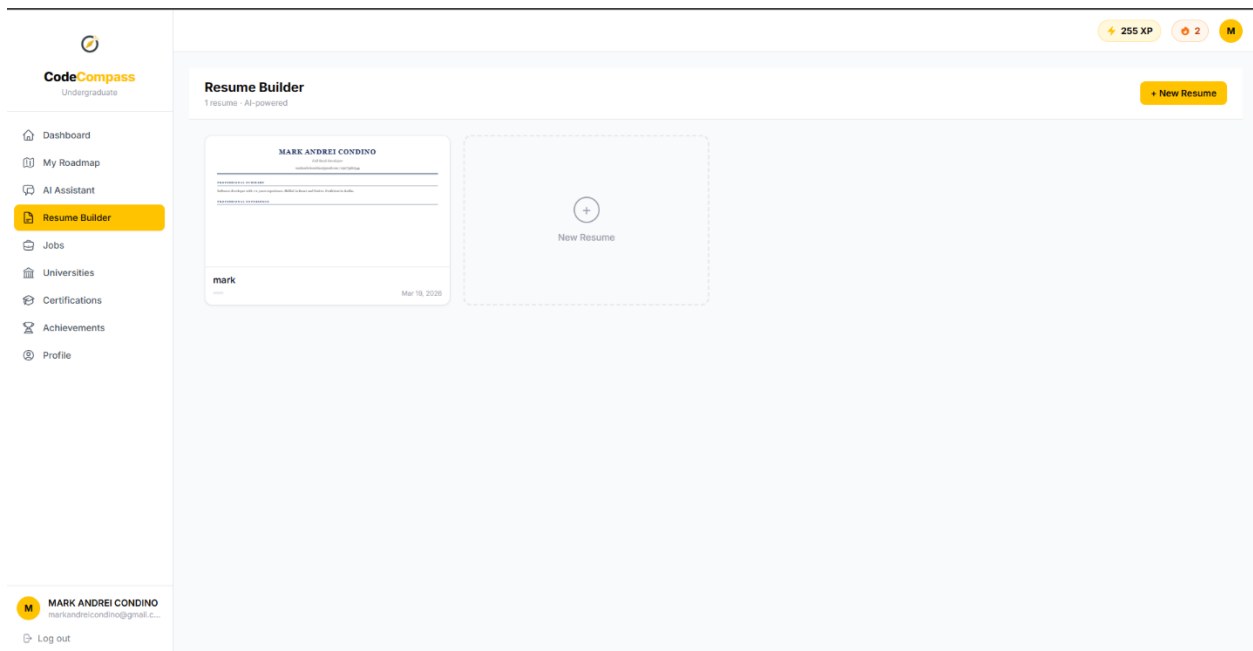
DASHBOARD



ROADMAP



AI ASSISTANT



RESUME BUILDER

CodeCompass

Undergraduate

Dashboard

My Roadmap

AI Assistant

Resume Builder

Jobs

Universities

Certifications

Achievements

Profile

245 XP

1

M

Philippines Only

Job Board

IT jobs in the Philippines.

All Jobs

Saved

Upload your resume PDF to get personalized job matches

PDF only

Search jobs... (e.g. React developer, data analyst)

All

Full Time

Part Time

Internship

Remote

Failed to load jobs.

Software Engineer II (Java)

REED ELSEVIER SHARED SERVICES (PHILIPPINES) INC.

9 Ninos, PH

Accountabilities • Design and develop data applications, REST APIs, and microservices (including middleware APIs) within a cloud-based, microservice-oriented architecture. ...

Today

View details --

Front End Dev (UI Engineer)

MicroSourcing

9 Manila, PH

Discover your 100% YOU with MicroSourcing! Position: Front-End Developer (UI Engineer) Work setup & shift: Work From Home | Nightshift Why join MicroSourcing?...

Today

View details --

Software Developer (Laravel & Livewire)

Intellect Philippines Branch Office

9 Pasig, PH

Responsibilities • Cloud-based Web application development using open-source technologies PHP Laravel, Livewire, InfluxDB, PostgreSQL Database, and AWS. • Play a...

Today

View details --

Software Engineer, Cloud - Sustaining Engineering

Canonical

9 Manila, PH

Canonical is a leading provider of open source software and operating systems to the global enterprise and technology markets. Our platform, Ubuntu, is very widely used in...

Today

View details --

Senior .NET Software Developer

Eastvantage Business Solutions Inc.

9 Taguig, PH

We are looking for a highly experienced and forward-thinking Senior .NET Software Developer to join our growing engineering team. This role is ideal for someone...

Today

View details --

Project-based Programmer

Land Bank of the Philippines (Official)

9 Manila, PH

The role involves migrating and converting legacy programs (COBOL, JCL, Easytrieve) to a new ZOS environment, customizing code, and performing comprehensive testing...

Today

View details --

MARC LAURENCE CON...

remconcepcion@lip.edu.ph

Log out

JOB RECOMMENDATION

CodeCompass

Undergraduate

Dashboard

My Roadmap

AI Assistant

Resume Builder

Jobs

Universities

Certifications

Achievements

Profile

245 XP

1

M

CHED CofC/CoD

University Finder

Find the right university for your IT/CS career in the Philippines.

Recommended for You

based on your profile & career goals

Mindanao State University – Iligan Institute of Technology

9 Iligan City, Region X

Offers BSIT CHED Center of Excellence Level 4 Accredited

BSCS BSIT

Free tuition (RA 10931)

View Details

Mapúa University

9 Manila, NCR

Offers BSIT CHED Center of Excellence Level 3 Accredited

BSCS BSCS BSIT

From ₱50,000/sem

View Details

Technological Institute of the Philippines

9 Manila, NCR

Offers BSIT CHED Center of Excellence Level 3 Accredited

BSCS BSCS BSIT

From ₱35,000/sem

View Details

University of the East

9 Manila, NCR

Offers BSIT CHED Center of Excellence Level 3 Accredited

BSCS BSIT

From ₱33,000/sem

View Details

Polytechnic University of the Philippines

9 Manila, NCR

Offers BSIT CHED Center of Development Level 3 Accredited

BSCS BSIT

Free tuition (RA 10931)

View Details

Technological University of the Philippines

9 Manila, NCR

Offers BSIT CHED Center of Development Level 2 Accredited

BSCS BSIT

Free tuition (RA 10931)

View Details

All Universities

Search by name, city, or region...

All Programs

All Types

36 universities found

AMA Computer University

9 Quezon City, NCR

Private

Adamson University

9 Manila, NCR

Private

MARC LAURENCE CON...

remconcepcion@lip.edu.ph

Log out

UNIVERSITIES PAGE

CodeCompass

undergraduate

Dashboard

My Roadmap

AI Assistant

Resume Builder

Jobs

Universities

Certifications

Achievements

Profile

M

MARC LAURENCE CON...

marcconception@tsp.edu.ph

Log out

245 XP

1

M

Certifications

Track free IT certifications that strengthen your resume.

Browse All

My Certs (0)

Q

Search certifications...

All

TESDA

Google

freeCodeCamp

CS50

IBM

Cisco

Microsoft

AWS

CompTIA

Fortinet

Linux Found.

MongoDB

Kaggle

HackerRank

GitHub

Meta

HubSpot

Postman

Scrum

10 certifications — click any card to see how to get it

A

AWS

AWS Educate — Cloud Computing Fundamentals

Available

Cloud / DevOps

Beginner

AWS Educate free badge: cloud concepts, AWS core services, storage, compute, and networking basics. No AWS account or credit card required. Earn a free Credly badge after completing the course path. ~5-8 hours.

0-8h

Free

Track This

Details

A

AWS

AWS Educate — Machine Learning Foundations

Available

Data Science / AI

Beginner

AWS Educate free badge: ML concepts, supervised/unsupervised learning, AWS ML services overview, and ethical AI. No AWS account or credit card. ~5 hours. Free Credly badge.

0-5h

Free

Track This

Details

A

AWS

AWS Knowledge: Cloud Essentials

Available

Cloud / DevOps

Beginner

AWS Knowledge Badge: cloud concepts, AWS core services, security model, and billing fundamentals. Pass an 80-question assessment to earn a free Credly badge. ~4-6 hours. No AWS account required.

0-5h

Free

Track This

Details

N

Cisco

Data Analytics Essentials

Available

Data Science / AI

Beginner

Intro to data analytics: collecting, cleaning, visualizing, and interpreting data. Covers basic statistics and real-world scenarios. Free digital badge on completion. ~30 hours.

0-30h

Free

Track This

Details

CERTIFICATION PAGE

CodeCompass

Undergraduate

Dashboard

My Roadmap

AI Assistant

Resume Builder

Jobs

Universities

Certifications

Achievements

Profile

M

MARC LAURENCE CON...

marcconception@tsp.edu.ph

Log out

245 XP

1

M

Achievements

Earn XP, collect badges, and compete with other students.

XP Progress

245

Total XP

Level 1 → Next level at 500 XP

Level 1

245 / 500 XP

Daily Streak

1

day streak

Great job! Keep logging in every day to maintain your streak.

Badges

0 / 8 earned

badge-check

Certified

Earn your first certification.

academic-cap

First Step

Complete your first roadmap node.

trophy

Node Master

Complete 10 roadmap nodes.

map

Road Builder

Generate your first learning roadmap.

chat-bubble

Onboarded

Complete the AI onboarding chat.

user-circle

fire

Week Warrior

1 out of 3 days on a streak

star

Consistency King

1 out of 30 days on a streak

Leaderboard

This Week

This Month

All Time

1

MARC LAURENCE CONCEPCION

245 XP

2

MARK ANDREI CONDINO

10 XP

3

MARK ANDREI CONDINO

10 XP

Complete nodes to earn XP and climb the board!

ACHIEVEMENT PAGE


CodeCompass
Undergraduate

Dashboard

My Roadmap

AI Assistant

Resume Builder

Jobs

Universities

Certifications

Achievements

Profile

 **MARC LAURENCE CON...**
mmiconcepcion@itp.edu.ph

Log out

245 XP

1



My Profile

Edit Profile

 **MARC LAURENCE CONCEPCION**
mmiconcepcion@itp.edu.ph
Undergraduate Student

Academic Info

Year Level

Select year level

Program

BSIT

Target Career Path

Select target career

Bio

Tell us a bit about yourself...

LinkedIn URL

https://linkedin.com/in/...

GitHub URL

https://github.com/...

Current Skills

Edit your profile to update your skills.

Python

JavaScript

React

Django

Node.js

Java

C++

SQL

HTML/CSS

Git

Machine Learning

Data Analysis

Networking

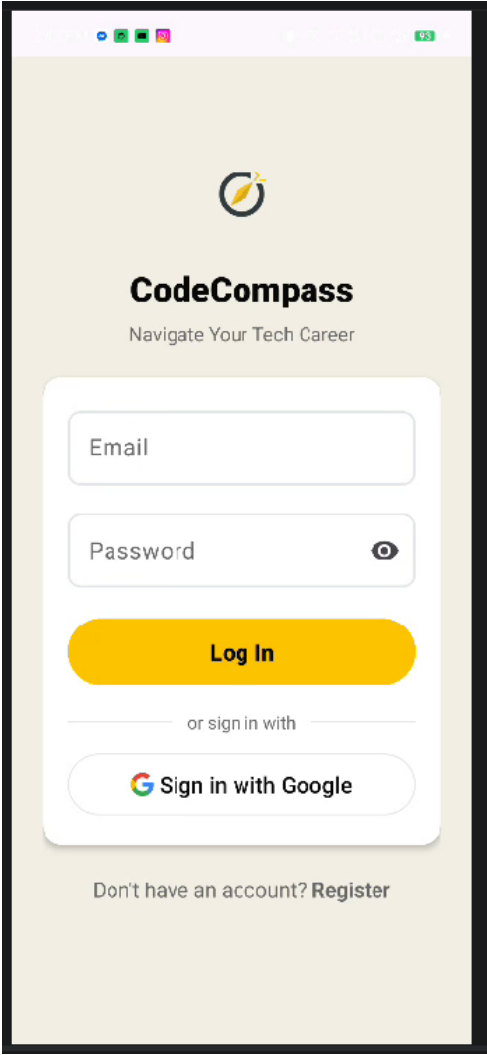
Linux

Flutter (AWS/GCP)

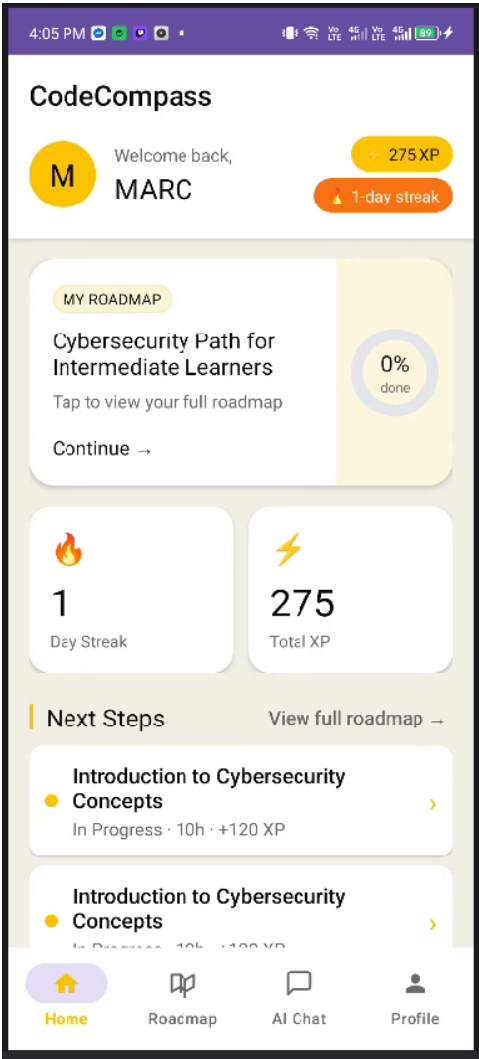
Mobile Development

PROFILE PAGE

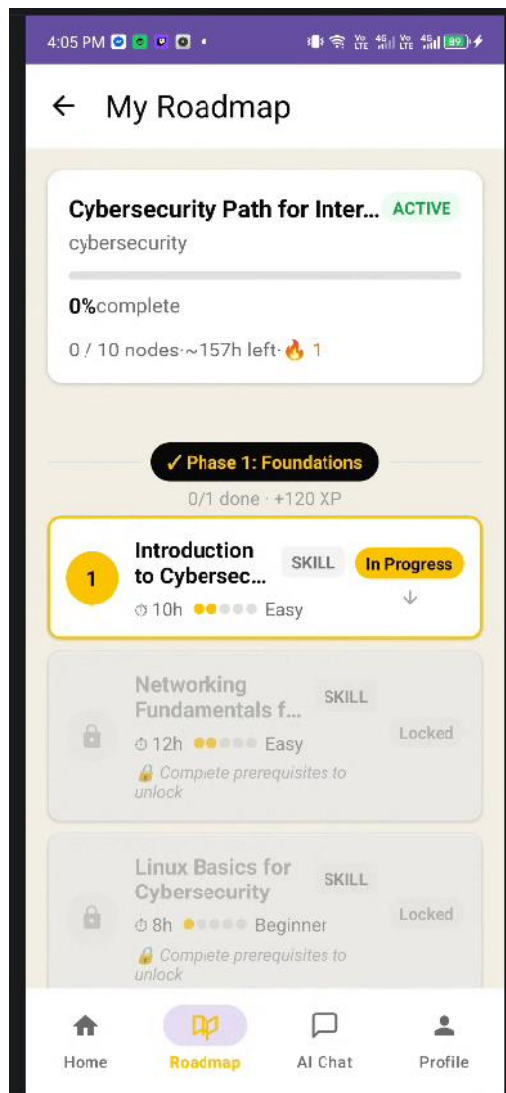
ANDROID VERSION



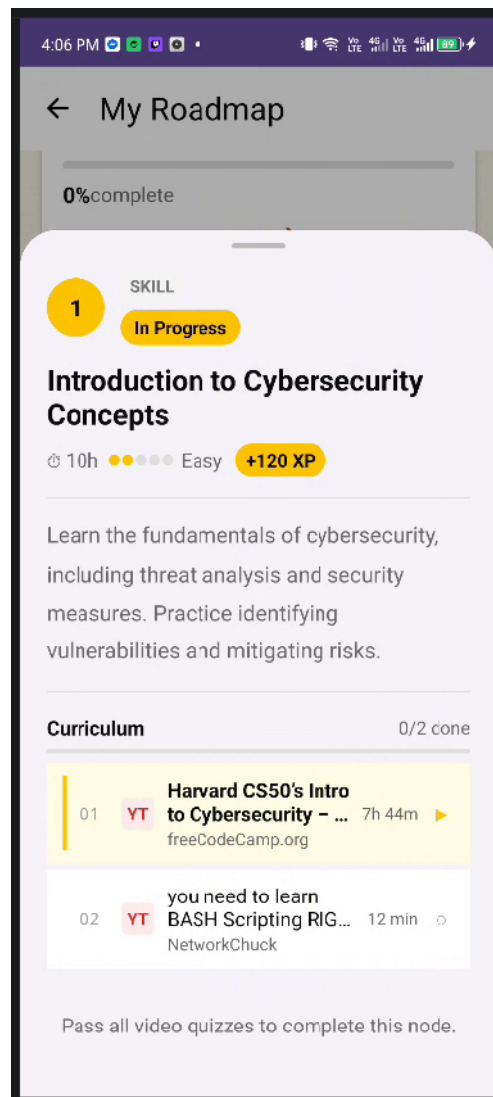
LOGIN



DASHBOARD



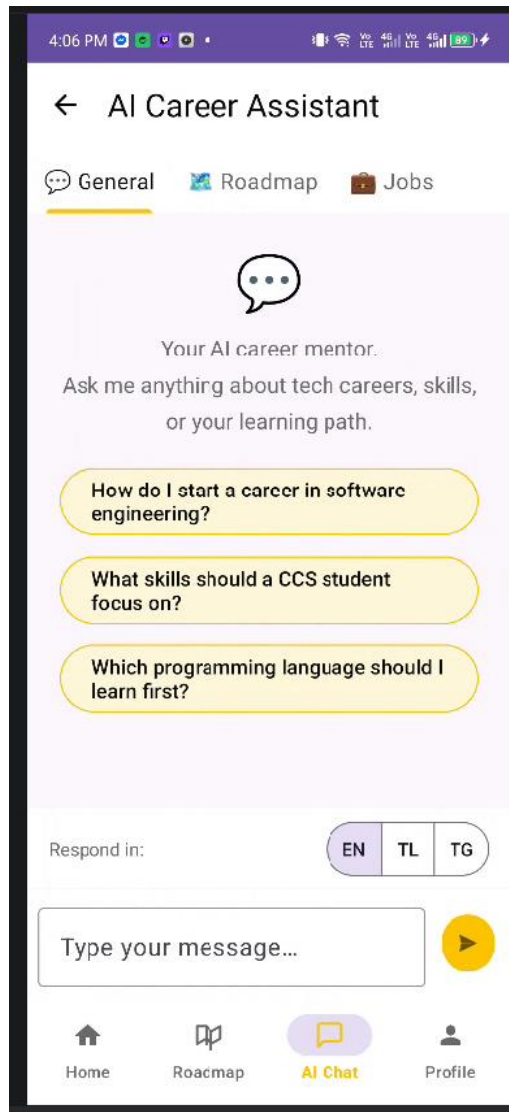
ROADMAP



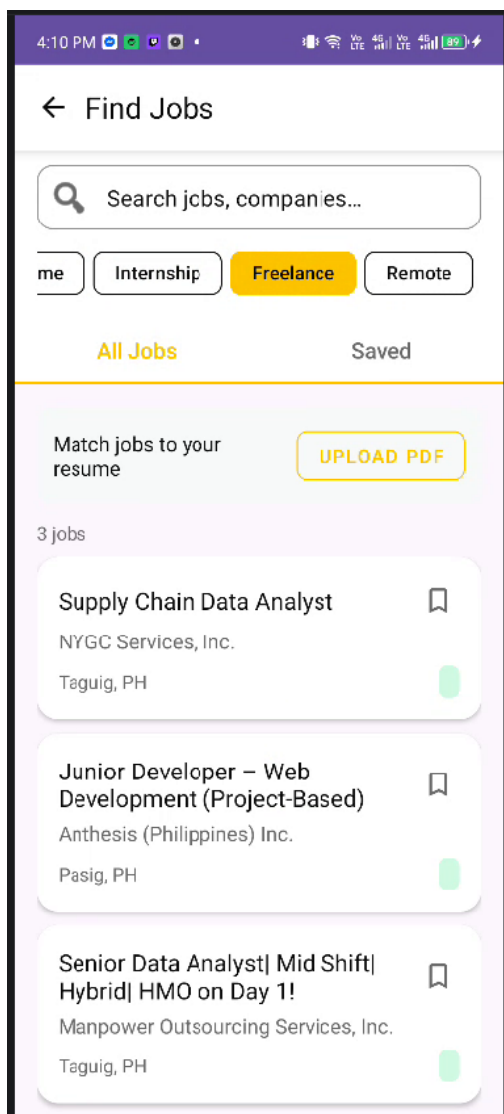
MODULE PAGE



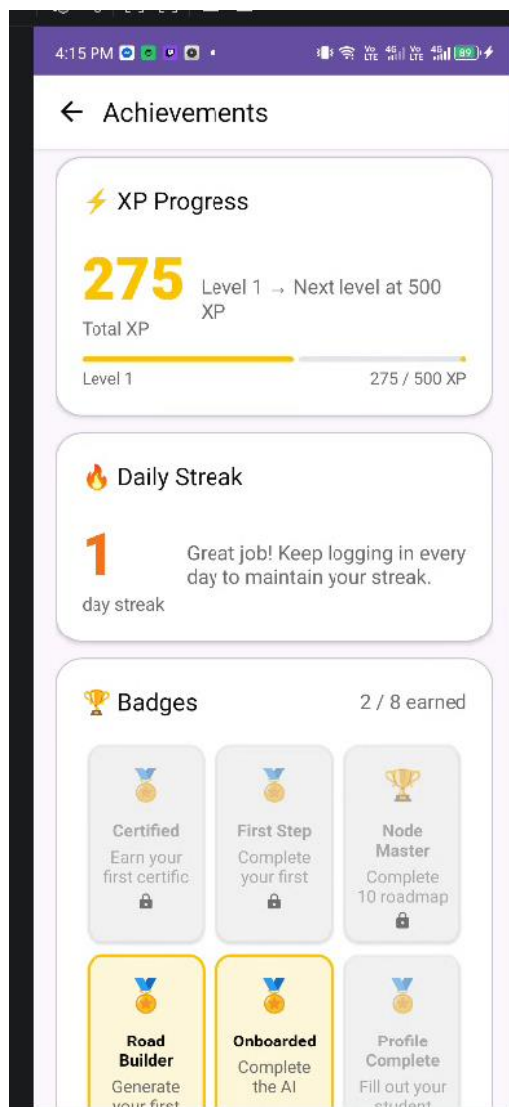
DETAILED LESSON PAGE



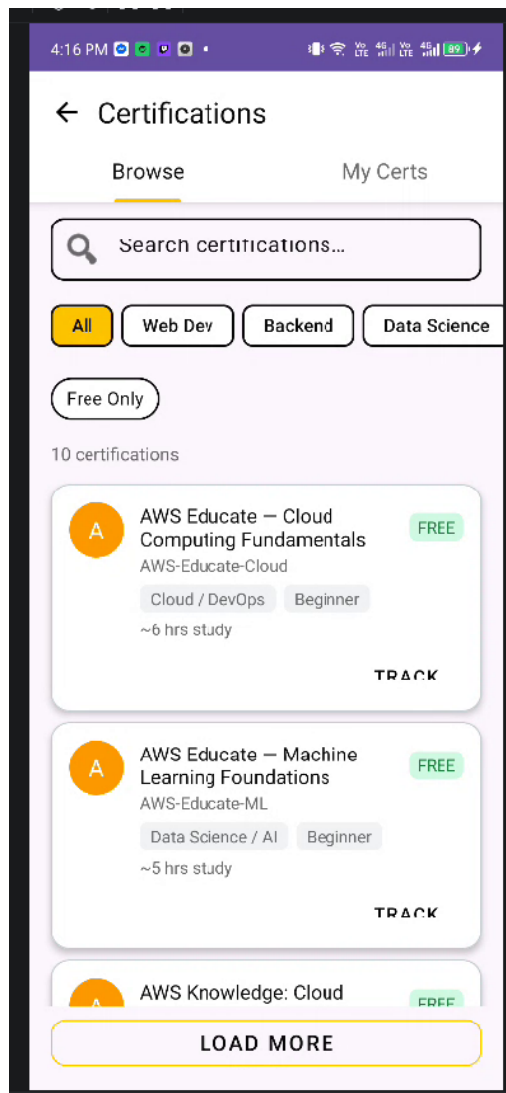
AI ASSISTANT



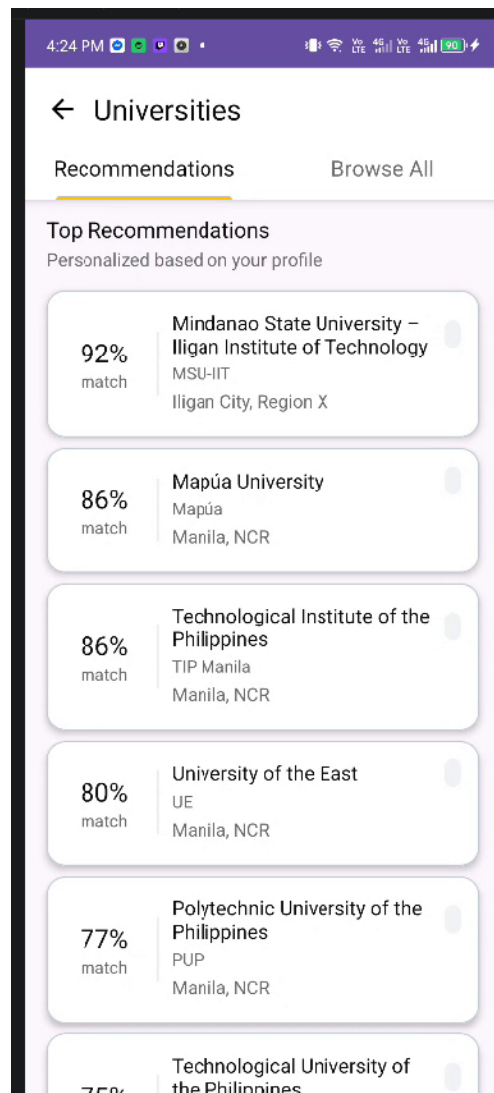
JOBS PAGE



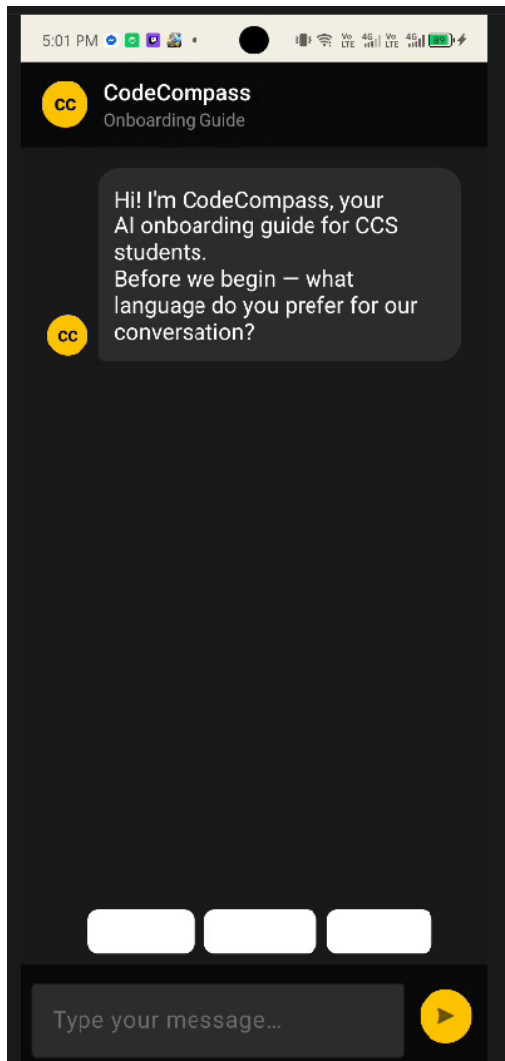
ACHIVEMENTS



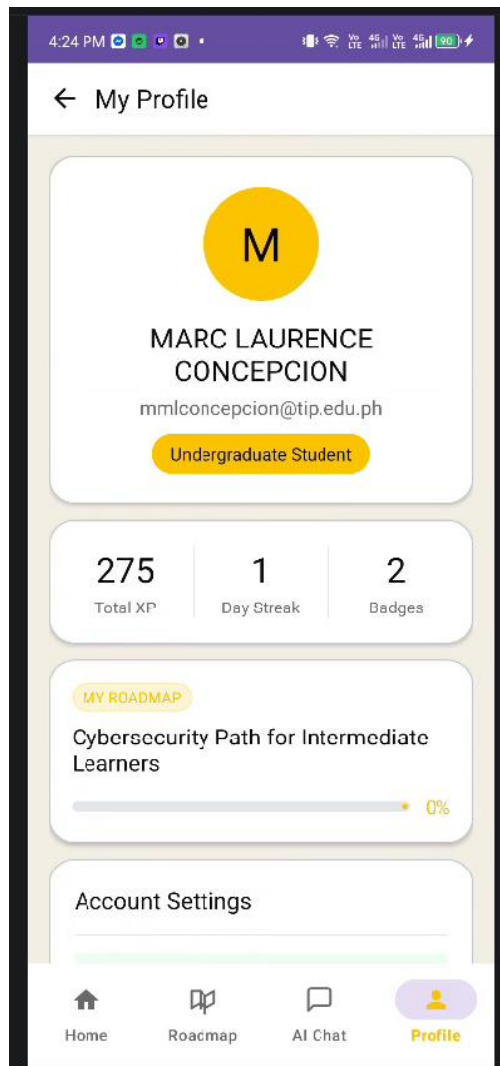
CERTIFICATIONS



UNIVERSITIES



ONBOARDING



PROFILE